



Andrey Yershov

Senior / Lead Systems Engineer

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Summary

Multidisciplinary systems engineer with 7+ years of experience combining photonics, machine vision, automation software, hardware integration, and customer deployment. I turn integrative research and customer needs into reliable intelligent optical and robotic systems – from feasibility and architecture through integration, deployment, and technical leadership.

Professional Experience

July 2023 – present **Project Lead, Custom Inspection Systems**, Dioptic GmbH – Weinheim, Germany

- Led development of a white-light-interferometry system for internal lens-assembly measurements with 70 nm accuracy, owning concept, calibration, feasibility, component selection, Python/asyncio backend, and TypeScript/Vue frontend for a EUR 190k project.
- Led feasibility studies for 100x wafer-microscopy autofocus and in-line medical-device camera inspection, reducing refocus time from more than 100 ms to 7 ms and inspection cycle time from 80 s to 3 s.
- Led internal development of a full-field optical coherence tomography device for lens subsurface-damage inspection, including real-time multithreaded acquisition, Numba processing, 3D scatterer detection, and napari reconstruction.
- Built a FastAPI robot-control backend deployed across 5 cleanroom workcells in a EUR 7M+ program, translating HTTP/WebSocket commands into validated motion jobs, telemetry, safety diagnostics, and simulation or live modes; supported deployment in South-East Asia across 15+ visits.
- Built a real-time Universal Robots control package with RTDE state streaming, TCP/IP communication, multithreaded controller interfaces, and shared state for pose, joints, I/O, runtime status, and safety diagnostics.
- Designed software and network architecture covering state machines, typed APIs, external configuration, offline/containerized deployment, QA tooling, packaging, network segmentation, routing, PoE management, and device communication.
- Developed a trimesh digital twin for collision checking in a 9-axis workcell and produced fixtures with Autodesk Fusion and FDM/SLA 3D printing.

Aug 2022 – June 2023 **Computer Vision Software Developer**, Xsight Optics GmbH – Jena, Germany

- Developed a real-time face-detection and skin-extraction pipeline for a medical product prototype, allowing clinical testing involving 100+ patients to proceed.
- Improved patient face-detection accuracy from 63% to 92% in challenging clinical scenarios, outperforming the previous outsourced solution.
- Developed real-time rPPG and contactless heart-rate estimation on Jetson/Linux, combining MediaPipe landmarks, multispectral processing, physiological-signal denoising, and moving-window spectral analysis.

- Mar 2022 – July 2023 **Photonics Research Associate**, Leibniz Institute of Photonic Technology – Jena, Germany
- Built transformer-based semantic-segmentation models with SegFormer and ViTAE/UPerNet for subcellular biological-structure multiplexing and tracking, achieving 95%+ precision on real data.
 - Applied transfer learning from a comparable non-biological modality to address limited biological training data, reducing fine-tuning time by 87% from 24 h to 3 h.
 - Created a Single-Molecule Localization Microscopy framework for a neuromorphic event camera, improving computed image resolution from 65 nm to 27 nm and enabling high-temporal-resolution analysis of biological dynamics.
- May 2019 – Sept 2021 **Undergraduate Robotics Researcher**, Nazarbayev University – Astana, Kazakhstan
- Implemented a deep-learning model for video action recognition with 97%+ accuracy on a 7-class dataset, followed by an iteration for age and gender estimation from actions.
 - Evaluated Temporal Segment Networks and SlowFast Networks using optical flow, confusion-matrix analysis, architecture ablation, and grid-search tuning.
 - Developed an educational reinforcement-learning setup with a snake-like robot on ROS2 that was used for teaching 150+ undergraduate students.
- June 2015 – Sept 2021 **Founder / B2B Industrial Photographer**, androbaza – Astana, Kazakhstan
- Founded and ran a B2B industrial photography business, delivering assignments for National Geographic, Forbes, universities, and IT hubs while managing customer requirements, deadlines, editing, and stakeholder-facing delivery.
 - Created technical product photography, infrared-wavelength product photography, and visual material in complex industrial environments; recognized as an international top-100 photographer in 2019 and a multiple-time national photography award winner.
- Sept 2018 – Apr 2021 **Regional Technical Manager**, eduvation.kz – Astana, Kazakhstan
- Managed a 5-person regional sales team and a USD 50k operating budget for educational course programs, increasing monthly active students from 60 to 450.
- Aug 2016 – Sept 2018 **Electronics Repair Specialist**, Apple Repair Center – Astana, Kazakhstan
- Performed laptop and electronics repair services alongside undergraduate studies.

Education

- Jan 2021 – Jan 2023 **Abbe Center of Photonics**, MSc in Photonics – Jena, Germany
Research-focused photonics MSc; DAAD scholarship; top 3%; thesis built and programmed an event-camera super-resolution microscope with temporal resolution down to 1 us.
- Jan 2017 – Jan 2021 **Nazarbayev University**, BSc in Robotics and Mechatronics – Astana, Kazakhstan
Full scholarship; president's stipend; top 5%; thesis built a data-driven DC-motor control pipeline using Arduino/STM32, MATLAB/Simulink, system identification, and real-time closed-loop testing.

Selected Publications

- Jan 2025 **ChildACT: Child Action Recognition Dataset in RGB Data**
ACM HRI '25
- Jan 2025 **NanTex enables computational multiplexing and phenotyping of organelles across super-resolution modalities**
bioRxiv (Contribution)

- Jan 2023 **Event-based vision sensor for fast and dense single-molecule localization microscopy**
Nature Photonics (Contribution)
- Jan 2023 **Handgerät für die kontaktlose Messung von Vitalparametern und Verfahren zur Messung von Vitalparametern**
WO2025051856A1; published application; patent application (Contribution)

Core Skills

Software Engineering, APIs & Data: Python, asyncio, FastAPI, Pydantic, WebSocket, OpenAPI, TypeScript, Vue, NumPy, Numba, multiprocessing, multithreading, Docker, pytest, GitLab CI, Linux, Bash, systemd, configuration, logging, monitoring, technical documentation

Computer Vision, Machine Learning & Bioimaging: OpenCV, PyTorch, scikit-learn, semantic segmentation, SegFormer, ViTAE/UPerNet, transfer learning, MediaPipe, rPPG, multispectral processing, action recognition, optical flow, GStreamer, Jetson edge AI, 3D vision, point clouds, coordinate transformations, fiducial registration

Photonics, Optical Metrology & Imaging: White-light interferometry, OCT, interferometric autofocus, SMLM, event-camera microscopy, optical metrology, spectroscopy, telecentric optics, optomechanical alignment, camera calibration, hardware triggering, beam shaping, piezo motors, closed-loop control, napari, basic Zemax usage

Robotics, Automation & Industrial Systems: Universal Robots, RTDE, URScript, ROS2, robot-state abstraction, safety diagnostics, runtime state machines, digital twins, trimesh collision checking, PLC interfaces, Modbus TCP, PROFINET, EtherCAT, industrial Ethernet, cleanroom commissioning, FMEA, FAT/SAT, familiar with ISO 10218 and ISO 15066

Embedded Hardware & Prototyping: Microcontrollers, STM32, Arduino, embedded C, FPGA fundamentals, PWM motor actuation, DC motor drivers, MATLAB/Simulink, system identification, Autodesk Fusion, FDM/SLA 3D printing, PixHawk, UAV development, electronics repair

Technical Leadership & Delivery: Project leadership, feasibility studies, requirements engineering, customer-facing development, product ownership, budget and team management, customer-site deployment, design controls, requirements traceability, risk management, verification and validation, familiar with ISO 13485 and IEC 62366-1

Selected Projects

- Jan 2022 **Search & Rescue Octocopter (UAV) Development – Jena, Germany**
Designed and constructed a modular drone with 10 kg payload using the open-source PixHawk autopilot with EUR 20k funding.
- 2018-2023 **Open Source Contributor**
Contributed to repositories of OpenMMLab, Robotics Group of University of Zurich, and Julia Programming Language.
- Jan 2026 **Local LLM Setup**
Set up and maintained local LLM and VLAM environments for offline experimentation and practical AI tooling workflows.

Languages

Proficiency: English (C1 / Full working), German (B2 / Working), Russian and Kazakh (Native / Bilingual), Turkish (Limited Working)